SIMATS ENGINEERING

ASSESSMENT TOOL 2 : Technical Presentation

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

COURSE OUTCOME COVERED

CO1: *Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions.* (BL2)

Dave's Taxonomy: Imitation (Level 1) **Total Marks: 50**
Question Count: 5

Code of Conduct

I Nirmal Kasi 192424248 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Nirmal

Assessment Tasks

Technical Presentation Topics

Prepare a 6-8 minute presentation on any one of the following topics. Your slides must include definitions, architecture, industry relevance, and one practical example.

Here are 10 additional technical presentation topics in a similar style, suitable for a 6–8 minute presentation with definitions, architecture, industry relevance, and a practical example:

1. **Containerization and Cloud-Native Applications: Transforming Modern Software Deployment**
   - Explore containers, container architecture, benefits, and a real-world deployment example.
2. **Serverless Computing: Reducing Infrastructure Management in the Cloud**
   - Discuss Function-as-a-Service (FaaS), architecture, business advantages, and practical use cases.
3. **Edge Computing vs Cloud Computing: Choosing the Right Computing Paradigm**
   - Compare architectures, applications, and industry adoption scenarios.
4. **Microservices Architecture in Cloud Environments**
   - Explain microservices design, benefits over monolithic systems, and industry implementations.
5. **Cloud Security Fundamentals: Protecting Data and Applications**
   - Cover security models, shared responsibility architecture, and real-world security practices.
6. **Multi-Cloud and Hybrid Cloud Strategies for Enterprise IT**
   - Analyze deployment models, business benefits, challenges, and practical examples.
7. **The Role of Artificial Intelligence and Machine Learning in Cloud Platforms**
   - Discuss AI/ML cloud services, architecture, and industry applications.
8. **DevOps and Continuous Integration/Continuous Deployment (CI/CD) in Cloud Computing**
   - Explain DevOps principles, CI/CD architecture, and practical implementation examples.
9. **Cloud Storage Technologies: Object, Block, and File Storage Compared**

    - Compare storage architectures, use cases, and industry adoption patterns.
10. **Internet of Things (IoT) and Cloud Computing: Building Connected Ecosystems**
    - Examine IoT-cloud architecture, benefits, challenges, and real-world applications.
11. **Cloud Migration Strategies: Moving Enterprise Applications to the Cloud**
    - Examine migration approaches, cloud architecture considerations, benefits, challenges, and a successful migration case study.
12. **Big Data Analytics in Cloud Computing**
    - Explore cloud-based data processing architectures, analytics platforms, business value, and real-world applications.
13. **Identity and Access Management (IAM) in Cloud Environments**
    - Discuss authentication, authorization architectures, security benefits, and practical enterprise implementations.
14. **Content Delivery Networks (CDNs) and Their Role in Cloud Services**
    - Explain CDN architecture, performance optimization, industry relevance, and examples from popular web platforms.

Minimum slide flow:

- Title and objective
- Core technical explanation
- Architecture or process diagram
- Real-world use case
- Conclusion and key takeaway
- Limitations and Future
- Scope

CDN Architecture, Performance Optimization, Industry Relevance, and Examples

Presented by:

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N Gowri Sankar (192411246)
CSA1507 :CLOUD COMPUTING AND BIG DATA ANALYTICS
Date: 10/06/2026

Technical Presentation

Introduction

A **Content Delivery Network (CDN)** is a geographically distributed network of **edge servers** that delivers web content from locations closer to end users. By caching and serving resources such as **HTML, CSS, JavaScript, images, and videos** at the edge, CDNs reduce **network latency, bandwidth consumption, and origin-server load**. Modern CDNs also use technologies such as **Gzip/Brotli compression, HTTP/2, HTTP/3, QUIC, and TLS** to provide faster, secure, and reliable content delivery at scale.

Core Technical Explanation

Caching: Reduces repeated requests to the origin server.

Gzip/Brotli: Compress HTTP payloads to reduce bandwidth usage.

HTTP/2: Enables **multiplexing** and header compression over TCP.

HTTP/3 + QUIC: Uses UDP-based QUIC for lower latency and better packet-loss handling.

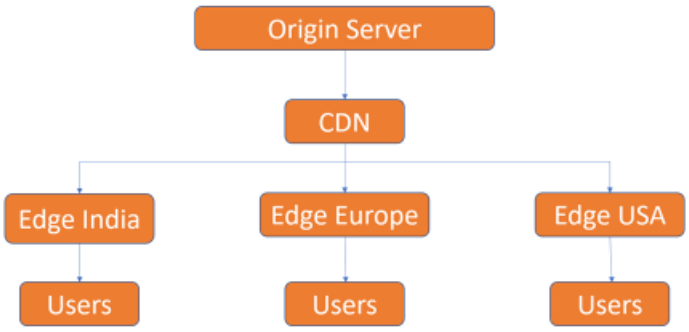
TLS: Provides encryption, authentication, and data integrity.

Load Balancing: Distributes traffic across edge/origin infrastructure.

Result: Lower **latency**, reduced **server load**, improved **throughput**, and higher **scalability and availability**.

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Architecture



Benefits

Lower Latency: Delivers content from geographically closer **edge servers**.

Faster Loading: Reduces **response time** and improves page-load performance.

Reduced Server Load: Caching minimizes requests to the **origin server**.

Scalability: Handles large traffic volumes and sudden **traffic spikes** efficiently.

Bandwidth Optimization: Compression and caching reduce **bandwidth consumption**.

High Availability: Distributed infrastructure provides **fault tolerance** and redundancy.

Improved Security: Provides **TLS encryption, DDoS protection, and WAF** capabilities.

Global Reach: Enables consistent and reliable content delivery to users worldwide.

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Real-World Applications

•**Netflix:** Uses distributed edge infrastructure to deliver high-quality video with low buffering.
•**YouTube:** Distributes videos and static assets through geographically distributed infrastructure.
•**Amazon CloudFront:** Delivers websites, APIs, images, videos, and software with low latency.
•**Cloudflare:** Provides CDN services along with **DDoS protection, WAF, DNS, and edge security**.
•**Akamai:** Uses a global edge network for large-scale **content delivery, web applications, and cybersecurity**.
•**E-Commerce:** Accelerates product images, JavaScript, CSS, and other static resources during high-traffic events.

Example Use Case

Scenario: An online shopping website receives thousands of users from different regions.

With CDN:

User → **Nearest Edge Server** → Cached Content

Product images, CSS, JavaScript, and videos are cached at **edge servers**.

Cache hits deliver content directly to users.

Cache misses retrieve content from the origin and store it at the edge.

Gzip/Brotli reduces file sizes.

HTTP/2/HTTP/3 improves connection performance.

TLS secures data transmission.

Result: Faster page loading, lower latency, reduced origin-server load, and better performance during traffic spikes.

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Conclusion and Key Takeaways

- **CDNs bring content closer to users** through geographically distributed edge servers.
- **Caching and compression** reduce latency, bandwidth usage, and origin-server load.
- **HTTP/2, HTTP/3, QUIC, and TLS** improve speed, reliability, and security.
- CDNs provide **scalability, high availability, and DDoS protection** for modern web applications.
- Widely used by platforms such as **Netflix, YouTube, Amazon CloudFront, Cloudflare, and Akamai**.

Key Takeaway: A CDN makes web applications **faster, more scalable, reliable, and secure** by delivering content efficiently from the network edge.

Limitations

Cost: Large-scale CDN usage can increase infrastructure and data-transfer costs.

Cache Invalidation: Updating cached content can be complex and may cause **stale content**.

Configuration Complexity: Requires proper **caching rules, routing, and security configuration**.

Dynamic Content: Frequently changing or personalized content gains less benefit from caching.

Third-Party Dependency: CDN outages or configuration failures can affect content delivery.

Security Risks: Misconfiguration can expose cached sensitive data or create security vulnerabilities.

Thank You!!!


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Google

Technical Presentation Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Technical Content	30%	Content is highly accurate, relevant, and insightful	Content is correct with minor gaps	Basic content with limited depth	Content is inaccurate or superficial
Structure and Flow	20%	Excellent logical flow and slide organization	Good structure with minor issues	Some organization but weak flow	Disorganized presentation
Use of Visuals	15%	Visuals strongly support technical communication	Relevant visuals used well	Limited or unclear visuals	No meaningful visuals
Communication Skills	20%	Confident, clear, and engaging delivery	Clear delivery with minor hesitation	Moderate clarity but uneven delivery	Weak delivery and low clarity
Professionalism	15%	Well-prepared and audience-focused presentation	Mostly professional	Basic preparation visible	Poor preparation and professionalism